Math 9 Fall 2022 — Lecture #3 Worksheet
Corresponding Outcomes: EO1, EO5, GOMa1, GOMa2

This is an in-class worksheet. You do not need to turn in anything.

1. Write a function `alt(n)` that takes as input a positive integer n and as output returns a length n row vector containing alternating -2s and 2s. Make sure nothing extra gets displayed to the screen.

2. Write a function `mults(m,n)` that takes as input two positive integers $m < n$ and as output returns a row vector containing all multiples of m that are *strictly* between 0 and n. Make sure that nothing else gets printed to the screen.

3. Write a function `bigmult(m,n)` that takes an input two positive integers $m < n$ and as output returns the biggest multiple of m that is strictly less than n. (Hint. Use your function `mults`. Don't retype the code...instead call that function.)

4. Write a function `countmult(m,n)` that takes an input two positive integers $m < n$ and as output returns the number of multiples of m that are strictly between 0 and n. (Hint. Use your function mults. Don't retype the code...instead call that function.)

5. Write a script `sinapprox.m` that generates the following output. Something to notice:

   (a) It displays a plot of $y = \sin(x)$ in blue. This plot should use 1000 equally spaced x-values between -6 and 6.

   (b) It displays a plot corresponding to $y = x - \frac{1}{6}x^3$ in red.

   (c) The red plot uses sample points every half unit along the x-axis and indicates those points with bullet.

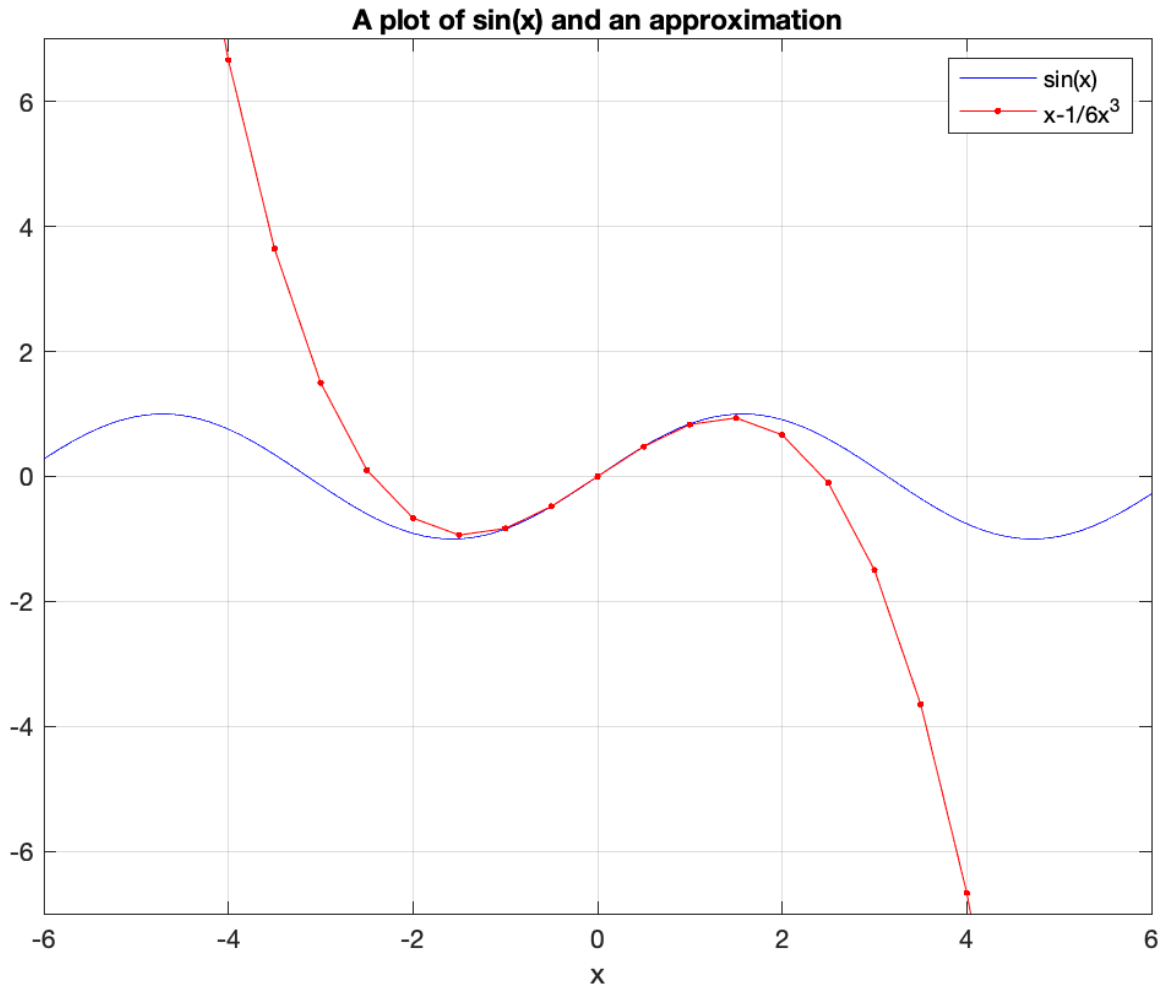
   (d) The border of the plot if $-6 \leq x \leq 6$ and $-5 \leq y \leq 5$.

   (e) The x-axis is labeled with an "x".

   (f) There are dashed grid lines. (Hint. Look up grid.)

   (g) There is a legend in the top right corner.

   (h) There is a title at the top of the plot.


A plot of sin(x) and an approximation
sin(x)
x-1/6x^3
6
4
2
0
-2
-4
-6
-6
-4
-2
0
2
4
6
x